$$\boldsymbol{p}_T = \begin{bmatrix} x_T \\ y_T \\ z_T \end{bmatrix}, \qquad {}^0R_T = \begin{bmatrix} n_x & o_x & a_x \\ n_y & o_y & a_y \\ n_z & o_z & a_z \end{bmatrix}, \qquad \boldsymbol{a} = \begin{bmatrix} a_x \\ a_y \\ a_z \end{bmatrix}$$

$$L_1 = 89.45 \text{ mm}, \qquad L_2 = 100 \text{ mm}, \qquad L_m = 35 \text{ mm},$$
$$L_3 = 100 \text{ mm}, \qquad L_4 = 129.15 \text{ mm}.$$

$$q_1 = \operatorname{atan2}(y_T, x_T)$$

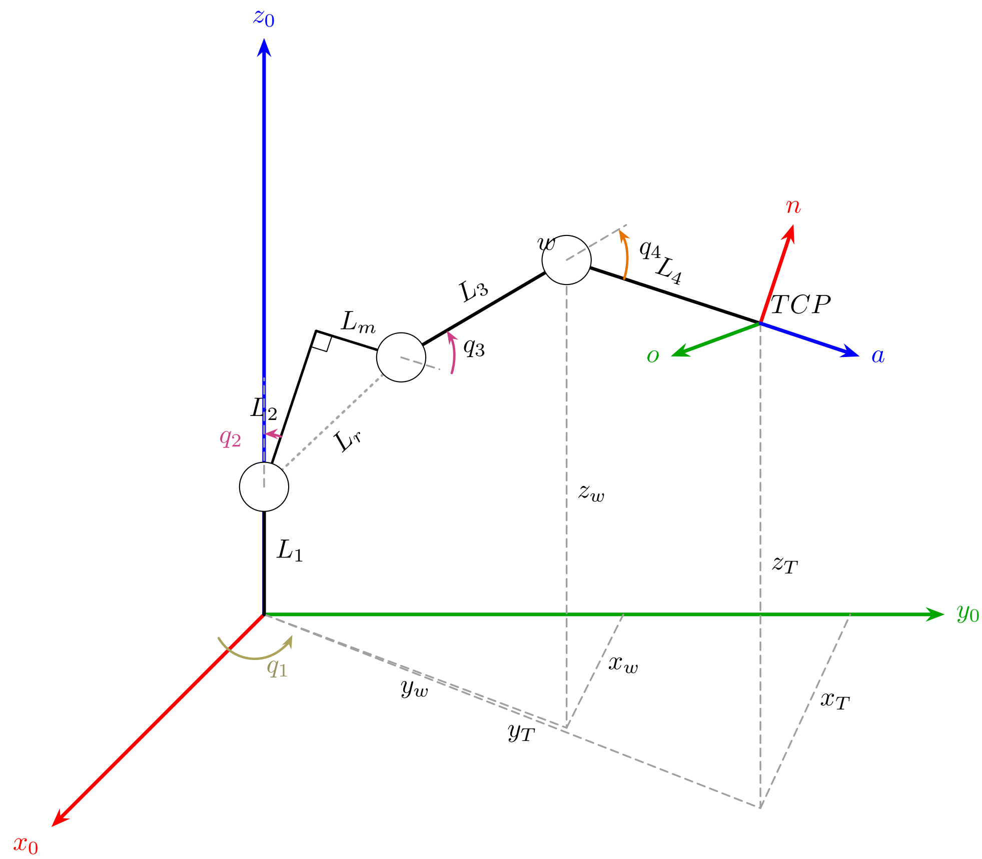
$$\theta_a = \operatorname{atan2}\left(\sqrt{a_x^2 + a_y^2}, a_z\right)$$

$$\theta_a = \theta, \qquad \boldsymbol{a} = \begin{bmatrix} \sin\theta\cos q_1 \\ \sin\theta\sin q_1 \\ \cos\theta \end{bmatrix}$$

$$\boldsymbol{p}_w = \boldsymbol{p}_T - L_4\boldsymbol{a}$$

$$\begin{bmatrix} x_w \\ y_w \\ z_w \end{bmatrix} = \begin{bmatrix} x_T \\ y_T \\ z_T \end{bmatrix} - L_4 \begin{bmatrix} a_x \\ a_y \\ a_z \end{bmatrix}$$

$$\boxed{\begin{aligned} x_w &= x_T - L_4\sin\theta\cos q_1, \\ y_w &= y_T - L_4\sin\theta\sin q_1, \\ z_w &= z_T - L_4\cos\theta. \end{aligned}}$$



z_0
n
w
q_4
L_4
L_3
TCP
L_m
q_3
o
a
L_2
q_2
L_r
z_w
L_1
z_T
y_0
q_1
x_w
y_w
x_T
y_T
x_0

$$L_r = \sqrt{L_2^2 + L_m^2} = \sqrt{100^2 + 35^2} = 105.9481 \text{ mm}$$

$$\beta = \text{atan2}(L_m, L_2) = \text{atan2}(35, 100) = 0.336675 \text{ rad} = 19.2900°$$

$$\psi = \frac{\pi}{2} - \beta = 1.234122 \text{ rad} = 70.7100°$$

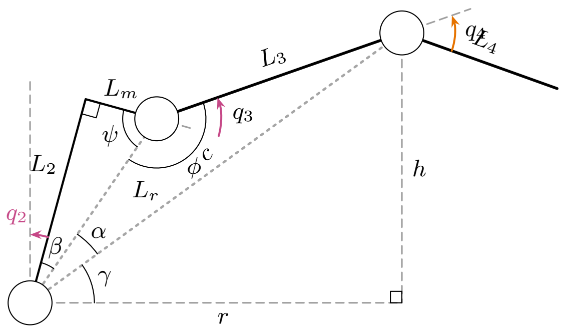
$$r = \sqrt{x_w^2 + y_w^2}, \qquad h = z_w - L_1, \qquad c = \sqrt{r^2 + h^2}$$

$$|L_r - L_3| \leq c \leq L_r + L_3$$

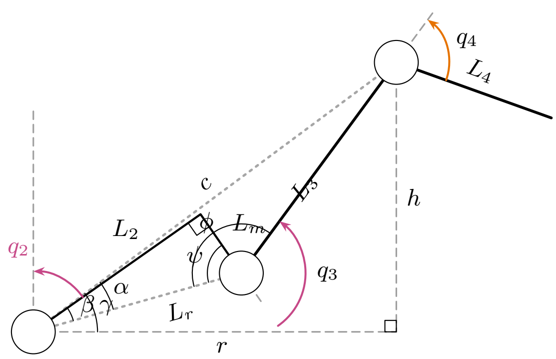
$$\phi = arccos\left(\frac{L_r^2 + L_3^2 - c^2}{2L_rL_3}\right) = \arccos\left(\frac{c^2 - L_3^2 - L_r^2}{-2L_rL_3}\right)$$

$$\gamma = \text{atan2}(h, r)$$

$$\alpha = arccos\left(\frac{L_r^2 + c^2 - L_3^2}{2L_rc}\right) = \arccos\left(\frac{L_3^2 - L_r^2 - c^2}{-2L_rc}\right)$$



Codo arriba



Codo abajo

$$
\boldsymbol{q}^{\uparrow} = \begin{bmatrix} q_1^{\uparrow} \\ q_2^{\uparrow} \\ q_3^{\uparrow} \\ q_4^{\uparrow} \end{bmatrix} = \begin{bmatrix} \operatorname{atan2}(y_T, x_T) \\ \dfrac{\pi}{2} - \beta - \alpha - \gamma \\ \pi - \psi - \phi \\ \theta_a - q_2^{\uparrow} - q_3^{\uparrow} - \dfrac{\pi}{2} \end{bmatrix}
$$

$$
\boldsymbol{q}^{\downarrow} = \begin{bmatrix} q_1^{\downarrow} \\ q_2^{\downarrow} \\ q_3^{\downarrow} \\ q_4^{\downarrow} \end{bmatrix} = \begin{bmatrix} \operatorname{atan2}(y_T, x_T) \\ \dfrac{\pi}{2} - (\gamma - \alpha + \beta) \\ -\pi + (\phi - \psi) \\ \theta_a - q_2^{\downarrow} - q_3^{\downarrow} - \dfrac{\pi}{2} \end{bmatrix}
$$

$$
\operatorname{wrap}_{[-\pi,\pi)}(q) = \operatorname{atan2}(\sin q, \cos q), \qquad \bar{\boldsymbol{q}}^{\,s} = \operatorname{wrap}_{[-\pi,\pi)}(\boldsymbol{q}^{s}), \qquad s \in \{\uparrow, \downarrow\}
$$

$$
\mathcal{S}_{\text{geom}} = \begin{cases} \{\boldsymbol{q}^{\uparrow}, \boldsymbol{q}^{\downarrow}\}, & |L_r - L_3| \le c \le L_r + L_3, \\ \varnothing, & \text{en otro caso} \end{cases}
$$

$$
\mathcal{S}_{\text{val}} = \{\bar{\boldsymbol{q}}^{,s} \in \mathcal{S}_{\text{geom}} : q_{i,\min} \le \bar{q}_i^{,s} \le q_{i,\max}, \quad i = 1, \dots, 4\}
$$

$$
d_s = \left\| \operatorname{wrap}_{[-\pi,\pi)}(\bar{\boldsymbol{q}}^{,s} - \boldsymbol{q}_{\text{actual}}) \right\|_2, \qquad s^* = \arg\min_{s:\, \bar{\boldsymbol{q}}^{,s} \in \mathcal{S}_{\text{val}}} d_s
$$

$$
\boldsymbol{q}_{\text{ejecutar}} = \begin{cases} \bar{\boldsymbol{q}}^{,s^*}, & \mathcal{S}_{\text{val}} \neq \varnothing, \\ \varnothing, & \mathcal{S}_{\text{val}} = \varnothing \end{cases}
$$

$$
\boxed{\begin{aligned}
&q_1 = \operatorname{atan2}(y_T, x_T), \\
&q_2^{\uparrow} = \frac{\pi}{2} - \beta - \alpha - \gamma, && q_2^{\downarrow} = \frac{\pi}{2} - (\gamma - \alpha + \beta), \\
&q_3^{\uparrow} = \pi - \psi - \phi, && q_3^{\downarrow} = -\pi + (\phi - \psi), \\
&q_4^{\uparrow} = \theta_a - q_2^{\uparrow} - q_3^{\uparrow} - \frac{\pi}{2}, && q_4^{\downarrow} = \theta_a - q_2^{\downarrow} - q_3^{\downarrow} - \frac{\pi}{2}.
\end{aligned}}
$$